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Inventor(s)	Hansen; Lars Hjorth et al.

Gripping system for an autonomous guided vehicle

Abstract

A gripping system for an autonomous guided vehicle (AGV) and such AGV are disclosed herein. The gripping system for automated gripping and pulling/pushing a cart comprises a unique gripping end effector ensuring controlled steering of the cart while allowing rolling of the cart relative to the body of the AGV. The end effector comprises means for indication of state of connection between the cart and the gripping system, ensuring a reliable, safe and efficient cart gripping and pulling operation.

Inventors:	Hansen; Lars Hjorth (Tommerup, DK), Pedersen; Mikkel Steen (Silkeborg, DK)
Applicant:	MOBILE INDUSTRIAL ROBOTS A/S (Odense SOE, DK)
Family ID:	71083388
Assignee:	MOBILE INDUSTRIAL ROBOTS A/S (Odense SOE, DK)
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Primary Examiner: Johnson; Kyle T

Attorney, Agent or Firm: ArentFox Schiff LLP

Background/Summary

(1) This application is a U.S. national stage entry of PCT application no. PCT/EP2021/064668 which was filed on Jun. 1, 2021. PCT application no. PCT/EP2021/064668 claims priority to European application no. EP20178999.7 which was filed on Jun. 9, 2020. This application claims priority to both PCT application no. PCT/EP2021/064668 and to European application no. EP20178999.7. Both PCT application no. PCT/EP2021/064668 and European application no. EP20178999.7 are incorporated into this this application by reference.

FIELD OF THE INVENTION

(2) The present invention relates to a gripping system for an autonomous guided vehicle for docking and undocking with a cart and for pulling and pushing the cart. More specifically the invention relates to a manipulator type gripping system.

BACKGROUND OF THE INVENTION

(3) Transportation of articles within a structure or between multiple structures has posed, to some degree, a problem. For example, hospitals with over 500 beds, need to use a considerable number of working hours on physical movement of linen, medical specimens, pharmaceuticals, blood products, patient charts, x-rays, and meals. Many transportation options exist today as

commercially available products, for example, conveyor belts, pneumatic tubes, gimbled carts, and mobile robots. Mobile robots and in particular Autonomous Guided Vehicles (AGV) have been equipped with a drive system under computer control, which allows autonomous guidance between multiple locations. Towing of carts with a rear mounted towing apparatus on AGV's are well known in the art. Commonly a commercial hitch and ball are mounted at one end of the AGV, usually the rear, whereat a cart comprising a releasable socket and a tow bar is connected. U.S. patent application Ser. No. 15/566,516 (publication number US2018/0281178) discloses a mobile robot system comprising an AGV and a gripping device mounted on the AGV. The AGV comprises a wheel-driving motor, an on-board computer, means for navigation, orientation and maneuvering in an environment with moving obstacles, a sensor system, and a wireless communication system for receiving and sending signals. The gripping system comprises a base for attaching the gripping system to the AGV, an arm and a gripper for gripping a cart. Main challenge for such gripping system is that position of the cart pulling AGV relative to the cart must be very precise in order to be able to grip the cart. This may require several attempts of docking and re-docking, and/or it is required that the docking be done at a slow speed. Another challenge of the prior art is that there is no safe indication of whether the gripper is in a correct alignment with to the connecting part of the cart before and/or while the gripping mechanism is activated. This can lead to an incorrect/weak coupling between the gripping device and the cart, and may lead to necessity of additional coupling attempts, increased wear and tear of the coupling mechanism, and to a situation where the cart is lost during transport.

(4) A further challenge with the prior art solutions is that there is no safe indication of whether a cart is correctly decoupled from the cart-pulling vehicle. This can lead to situations where a cart is accidentally still attached to the cart-pulling vehicle while the vehicle is in a “no-cart” driving mode and does not consider the extra foot print of the accidentally attached cart while navigating around obstacles and structural elements.

(5) It is the object of the present invention to improve upon the prior art and provide vehicle gripping system for an AGV with a safe, reliable and quick gripping mechanism and an AGV with such a system. A further object of the invention is to improve the possibility to grip a variety of carts, which may or may-not be specifically designed for automatic transport, in a smooth manner for a safe pulling or pushing operation. A further object is to provide a safer operation of the AGV with the gripper system when no cart is supposed to be attached.

SUMMARY OF THE INVENTION

(6) According to one aspect of the invention, a gripping system **1** for an autonomous guided vehicle (AGV) is disclosed. The gripping system **1** comprises a base **2** for attaching the gripping system **1** to a body **4** of the AGV, an arm **6** for manipulating an end effector **8** for gripping a cart for towing or pushing by the AGV. The arm **6** at one end is pivotally attached to the base **2**, allowing a yaw motion of the arm **6** relatively to the AGV. At the other end, the arm **6** is provided with an end effector **8** being attached to the arm **6**. Height of the end effector **8** is adjusted by the actuator **14** of the arm **6**.

(7) The end effector **8** is attached to the arm **6** via a pivoting connection **10**. This allows a roll motion while limiting a yaw motion of the end effector **8** relatively to the arm **6**. Two sensor arms **20a**, **20b** of the end effector **8** are adapted to cause the end effector **8** to pivot around its pivoting connection **10** if only one of the sensors arms **20a**, **20b** is pressed against a connecting part of the cart. The sensor arms **20a**, **20b** are further adapted to activate sensors **22a**, **22b** and when pressed against a connecting part of the cart, whereby signals are send to a control system of the gripping device, in order to indicate whether the end effector is in a correct position for the clamp force to be applied against the connecting part of the cart. The end effector **8** is provided with a hook **11** and a plurality of brackets **13a**, **13b** adapted for applying a clamping force on a part of the cart frame by moving the hook **11** in the direction of the brackets **13a**, **13b**. A further sensor activating element **11a** is attached to the hook **11** for activating a further sensor **23** attached to a front face **21**. When

the sensor **23** is activated, it sends a signal to the control system of the gripping device, indicating a fully contracted hook **11**. According to another aspect of the invention an autonomous guided vehicle comprising the gripping system **1** is disclosed. The autonomous guided vehicle comprises drive wheels, a robot body **4** mounted on said drive wheels, a control system utilizing a navigational system, and a gripping system **1** mounted on said robot body **4** for coupling the cart to said vehicle. The AGV further comprises at least one proximity sensor **4.1**, **4.2** mounted on the robot body **4**. The control system is coupled to said at least one proximity sensor **4.1**, **4.2** for adjusting calculated robotic position and detecting any obstacles.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Features of the invention believed to be novel and inventive are set forth with particularity in the appended claims. The invention itself, however, may be best understood by reference to the following detailed description of the invention, which describes exemplary embodiments, given in non-restrictive examples of the invention, taken in conjunction with the accompanying drawings, in which:

(2) FIG. **1** shows the gripping system according to the invention mounted on an AGV.

(3) FIG. **2** shows the end effector of the gripping system according to the invention.

(4) FIG. **3** shows the end effector of the gripping system according to the invention gripping an element of a cart frame.

(5) Preferred embodiments of the invention will be described herein below with reference to the drawings. Each figure contains the same numbering for the same or equivalent element.

DETAILED DESCRIPTION OF THE INVENTION

(6) Numerous specific details are presented in order to provide a complete and comprehensible description of the invention embodiment. However, the person skilled in art will understand that the embodiment examples do not limit the application of the invention which can be implemented without these specific instructions. Well-known methods, procedures and components have not been described in detail for the embodiment in order to avoid any misleading. Furthermore, this description should not be constraining the invention to the given embodiment examples but only be considered as one of possible implementations of the invention.

(7) FIG. **1** shows a gripping system **1** for an AGV. The gripping system **1** comprises a base **2**, an arm **6** and an end effector **8** with sensors **23**, **22a**, **22b**.

(8) The base **2** comprises means, for example bolt holes, for attaching the base **2** to the AGV, an electrical interface (not shown) for connecting the gripping system to the AGV for the AGV to be able to control the gripping system. The base **2** may comprise a section **5**, for example a recess or an opening, for interaction with the gripping system arm's **6** securing means **9** for securing the arm **6** in a fixed position when such position is needed. The base **2** further comprises means **7** for attaching the arm **6** to the base **2**, for example a vertical shaft. The arm's attachment means **7** is configured to receive the arm **6** for it to be able to at least partly swivel around the vertical axis of the attachment point i.e. yaw relatively to the body **4** of the AGV.

(9) The arm **6** further comprises a frame **16**. The frame **16** at one end is attached to the means **7** for attaching the arm **6** to the base **2**. Opposite end of the frame **16** has an end effector **8** attached thereto. The frame **16** may comprise an upper frame element **16a** and lower frame element **16b**. The upper frame element **16a** and/or the lower frame element **16b** at one end are attached to the means **7** for attaching the arm **6** to the base **2** and at the opposing end are attached to a sliding bar **16c**. The frame **16** is preferably rigid. The arm **6** further comprises an actuator **14** adapted to adjust the angle of a lifting bar **18a**, **18b** relative to the horizontal plane and thereby adjust height of a sliding element **17** as well as the height of the end effector **8**. The arm actuator **14** at one end is

fixed to the lower frame element **16b** and at another end is fixed to the lifting bar **18a** or a set of lifting bars **18a, 18b**. Each of the lifting bars **18a, 18b** are at their one end pivotally attached to a first end of respective support bars **19a, 19b** at attachment points **19c**. Each of the support bars **19a, 19b** at a second end pivotally attached to the lower frame element **16b** at attachment points **19d**. The lifting bars **18a, 18b** are at a respective second end pivotally attached to the sliding element **17**. The sliding element **17** is sliding up and down the sliding bar **16c** when the arm **6** actuator **14** acts upon the lifting bars **18a, 18b**. The sliding element **17** is connected to the end effector **8** via a pivoting connection **10**. The pivoting connection **10** allows the end effector **8** to partly roll relatively to the base **2** and the body **4** of the AGV.

(10) The arm **6** comprises securing means **9** for locking the arm **6** in a sideways stationary and stable position with respect to the body **4** of the AGV. The locking means **9** may be any electrically controllable mechanism. The securing means **9** preferably comprises a locking actuator **9a** and a pin **9b** which, when actuated protrudes from the frame **16** of the arm **6** towards the base **2** of the gripping system and into an opening **5** in the base **2** and/or in the frame of the AGV. The actuator **9a** is preferably secured at a first end to the upper frame element **16a** and at a second end to the lower frame element **16b**, and is preferably situated between lifting bars **18a, 18b**. In a preferred embodiment the actuator **9a** is secured to the lower frame element **16b** via a pin **9b** by inserting the pin **9b** into an opening **3** in the lower frame element **16b**. While the arm **6** is not locked by the locking means the pin **9b** remains inserted into the opening **3** in a retracted position. During locking of the arm **6**, the actuator **9a** extends the pin **9b** through the opening **3** and further into the opening **5** in the base **2** and/or in the frame of the AGV. When locked, the arm **6** is in parallel to the centerline of the body **4** of the AGV. During unlocking of the arm **6**, the actuator **9a** retracts the pin **9b** from the opening **5** in the base **2** and/or in the frame of the AGV.

(11) The arm **6** may further comprise protecting shields **30** on both sides of the arm **6**.

(12) The end effector **8** is connected to the sliding element **17** via a pivoting connection **10** for a substantially free pivoting motion at a pivoting connection **10**. The vertical position of the end effector **8**, relatively to the sliding bar **16c** is adjusted by sliding the sliding element **17** on the sliding bar **16c**. The end effector **8** further comprises a hook **11** for gripping a part of a cart frame and is attached to one end of an elongated element **11.1**. The elongated element **11.1** is movable linearly by actuation means **12**, such as a linear actuator, so that part of the cart frame when hooked is secured between the hook **11** and front face **21** of the end effector **8**. The actuation means **12** of the hook **11** preferably comprises a driving part **12.1** and a guiding part **12.2** for accommodating, moving and guiding the elongated element **11.1** for moving the hook **11**. The guiding part **12.2** is preferably disposed inside bottom part of a frame **15** of the end effector **8**. The driving part **12.1**, such as a motor for driving the actuator is preferably positioned near the sliding member **17** and upwards relatively from the guiding part **12.2**. The hook **11** further comprises a sensor activating element **11a**. The sensor activating element **11a** activates a sensor **23** positioned essentially on the front face **21** of the end effector **8**, approximately opposite the hook **11**. The sensor **11a** is activated whenever the hook **11** is moved into a completely closed position by the hook actuation means **12**. This ensure that the control system of AGV receives information that the hook **11** has moved into a completely closed position when there is no cart or other element accidentally attached to the hook **11**. Accordingly, the AGV can safely switch to a no-cart operation mode. Two supporting brackets **13a, 13b** are positioned laterally at the front face **21**, right and left of the hook **11**, respectively. Together with the hook **11**, the supporting brackets **13a, 13b** provide a clamp force against a connecting part of the cart. The end effector **8** is further provided with side sensors **22a, 22b** positioned in the frame **15** of the end effector **8**. The side sensors **22a, 22b** are preferably pressure sensors. Each of the side sensors **22a, 22b** are positioned between the respective bracket **13a, 13b** and a centerline of linear motion of the hook **11**. Each of the side sensors **22a, 22b** are activated by a respective sensor arm **20a, 20b** pivotally attached under the frame **15** of the end effector **8** and positioned between a respective bracket **13a, 13b** and a centerline of linear motion of the hook **11**.

First end **20a.1**, **20b.1** of each sensor arm **20a**, **20b** is protruding from the front face **21** of the end effector **8** so that when the end effector **8** starts gripping a connecting part of the cart each sensor arm **20a**, **20b** is pressed downwards by the connecting part of the cart when the end effector **8** is lifted upwards against the connecting part of the cart. Pivoting motion of each sensor arm **20a**, **20b** lifts second end **20a.2**, **20b.2** of respective sensor arm **20a**, **20b** thereby activating the respective pressure sensor **22a**, **22b**. When the end effector **8** is entirely disengaged from the connecting part of the cart the activating elements **11a**, **20a**, **20b** of each of the end effector sensors **23**, **22a**, **22b** are disengaging from the sensors **23**, **22a**, **22b** thus indicating that the end effector **8** is disengaged from the cart frame.

(13) During engagement of the end effector **8** and the cart frame, the end effector **8** is lifted against the cart connecting part of the cart. If the plane of the brackets **13a**, **13b** is different from the plane of the connecting part of the cart, one of the two sensor arms **20a**, **20b** is first pressed against the connecting part of the cart. This causes the end effector **8** to pivot around its pivot point at the pivoting connection **10** and thereby to align with the plane of the connecting part of the frame of the cart until both sensor arms **20a**, **20b** are in touch with the connecting part of the frame of the cart and eventually both side sensors **22a**, **22b** are activated. The two side sensors **22a**, **22b** are adapted to send signals to the control system of the AGV and the control unit is adapted to stop further lifting of the end effector **8** when both side sensors **22a**, **22b** are activated.

(14) While the end effector **8** and the connecting part of the frame of the cart are aligned, the control-system sends a signal to the hook **11** actuation means **12** to move the hook **11** into a closing position and thereby pressing the connecting part of the cart against the brackets **13a**, **13b**.

(15) While being in the operational mode with a cart attached, the AGV and/or the cart may drive on an uneven surface and roll relatively to each other. Had the connection between the frame of the cart and the gripping system been stiff in all directions, there would have been a risk of twisting between the hook/brackets and the connecting part of the cart and thereby a risk of losing the cart and/or damaging the hook/brackets and/or the connecting part of the cart.

(16) On the other hand, had the connection between the gripping system and the frame of the cart been flexible in all directions (e.g. a ball coupling), the ability of precise steering of the cart had been lost.

(17) With the end effector **8** being pivotally connected to the sliding element **17** at the pivoting connection **10**, the cart and the body **4** of the AGV can partly roll relatively to each other without the mechanical connection between them being twisted or stressed.

(18) When the cart is attached to the gripping system **1** the arm **6** is preferably in its center position relative to the AGV. The control system of the AGV identifies the cart by help of a camera/sensor and has stored information about of height and position of the connecting part/frame of the identified cart. The control system adjusts the height of the end effector **8** so that the hook **11** is at a height, which is low enough to let the hook **11** pass under the cart frame/attachment part. The AGV drives itself to a position in front of the cart so that the end effector **8** is close to the frame/attachment part. The hook **11** is driven outwards. The end effector **8** is lifted and the hook **11** is driven inwards until it together with the brackets **13a**, **13b** provides a gripping force on the cart frame.

(19) The construction ensures that the substantially same pulling forces can be transferred at different heights of the gripping portion, so that different type of cart can be pulled. Further the risk of tipping of the AGV is minimized.

(20) Another aspect of the invention is to provide an AGV with the gripping system **1** as disclosed above. The AGV for towing or pushing a wheeled cart using the gripping system **1** according to the first aspect of the invention comprises drive wheels, a robot body **4** mounted on said drive wheels, a control system utilizing a navigational system. The AGV further comprises proximity sensors **4.1**, **4.2** mounted on the robot body **4** for 270 degrees observation around the robot body **4**. One proximity sensor **4.1** is positioned at the back of the robot body **4**, in a corner. The other proximity

sensor 4.2 is positioned in the front of the robot body 4 in a corner. The back part is where the cart is attached, and the front part is on the opposite side of the robot body 4. The control system is coupled to the proximity sensors 4.1, 4.2 for adjusting the calculated robotic position and for detecting any obstacles.

(21) In a particularly preferred embodiment of the present invention, the navigational system includes an automatic control and guide system for reaching a given target and at the same time avoid collision with the surroundings.

(22) Through the disclosure AGV stands for Autonomous Guided Vehicle.

(23) Although numerous characteristics and advantages together with structural details and features have been listed in the present description of the invention, the description is provided as an example fulfilment of the invention. Without departing from the principles of the invention, there may be changes in the details, especially in the form, size and layout, in accordance with most widely understood meanings of the concepts and definitions used in claims.

Claims

1. A gripping system for a vehicle, the gripping system comprising: a base for attachment to the vehicle; an arm for manipulating an end effector; the end effector for gripping a cart to be towed or to be pushed by the vehicle, the end effector being attached to the arm by a pivoting connection configured to enable a roll motion of the end effector relative to the arm and to limit a yaw motion of the end effector relative to the arm, and the end effector comprising a first sensor arm and a second sensor arm, the first sensor arm for contacting the cart, the end effector being configured to move in response to the first sensor arm contacting the cart until the second sensor arm contacts the cart; wherein the arm comprises a first end and a second end, the first end being pivotally attached to the base to enable yaw motion of the arm relative to the vehicle, and the second end being connected to the end effector; and an actuator configured to adjust a height of the end effector.
2. The gripping system of claim 1, where the end effector is configured so that, when the first sensor arm contacts a connecting part of a frame of the cart, the end effector pivots around the pivoting connection to the arm until the second sensor arm contacts the connecting part of the frame of the cart.
3. The gripping system of claim 2, wherein the first sensor arm and the second sensor arm are configured to activate respective sensors to indicate that the end effector is in a correct position to apply a clamp force to the connecting part of the frame of the cart.
4. The gripping system of claim 3, wherein each of the first sensor arm and the second sensor arm comprises a first end protruding from a front face of the end effector and a second end for interacting with a respective sensor of the sensors.
5. The gripping system of claim 4, wherein the end effector comprises: a hook; and brackets configured to apply a clamp force on the connecting part of the frame of the cart by moving the hook in a direction towards the front face of the end effector using a hook actuator.
6. The gripping system of claim 5, wherein the end effector comprises a sensor, the sensor being responsive to an activating element attached to the hook to send a signal to a control system indicting a fully retracted hook.
7. The gripping system of claim 1, wherein the arm comprises securing means for restricting sideways motion of the arm; and wherein the base comprises a section for interacting with the securing means to secure the arm relative to the base.
8. The gripping system of claim 7, wherein the section of the base comprises a recess or an opening; wherein the securing means comprises a locking actuator and a pin; and wherein the recess or the opening of the section of the base is configured to receive the pin of the securing means.
9. The gripping system of claim 1, further comprising: a sliding element connected to the end

effector; wherein the arm comprises a rigid frame comprising: an upper frame element; a lower frame element; and a sliding bar connecting the upper frame element and the lower frame element, the sliding bar comprising a guiding element for adjusting the height of the end effector via the sliding element.

10. The gripping system of claim 9, wherein the arm comprises: lifting bars; and support bars; wherein a first end of each of the lifting bars is pivotally attached to a first end of a respective one of the support bars, a second end of each of the support bars is pivotally attached to the lower frame element, and a second end of each of the lifting bars is pivotally attached to the sliding element.

11. The gripping system of claim 10, wherein the actuator comprises a first end fixed to the lower frame element and a second end fixed to the lifting bars.

12. An autonomous guided vehicle (AGV) comprising: the gripping system of claim 1; wherein the vehicle comprises the AGV.

13. The AGV of claim 12, further comprising: a control system to control the actuator based on stored information about a height of a connecting part of a frame of the cart.

14. The AGV of claim 12, wherein the AGV comprises a mobile robot, the mobile robot comprising proximity sensors for detecting obstacles in an environment of the mobile robot.

15. The AGV of claim 14, wherein the mobile robot comprises a body having corners; and wherein the proximity sensors are on two corners of the body.

16. The AGV of claim 12, wherein the AGV comprises a mobile robot, the mobile robot comprising a navigation system, the navigation system comprising an automatic control and guide system for guiding the mobile robot to a target while avoiding collision with surroundings of the mobile robot.

17. The AGV of claim 12, further comprising: a sliding element connected to the end effector; wherein the arm comprises a rigid frame comprising: an upper frame element; a lower frame element; and a sliding bar connecting the upper frame element and the lower frame element, the sliding bar comprising a guiding element for adjusting the height of the end effector via the sliding element.

18. The gripping system of claim 1, wherein the vehicle comprises an autonomous guided vehicle.

19. The gripping system of claim 1, wherein each of the first sensor arm and the second sensor arm comprises an end protruding from a front face of the end effector for interaction with a frame of the cart.

20. The gripping system of claim 19, wherein the end effector comprises: a hook; and brackets configured to apply a clamp force on a connecting part of the frame of the cart by moving the hook in a direction of the front face of the end effector using a hook actuator.
